

Discrimination, Measurement, and Sorting

Labor I — Week 9

Teaching deck draft

Central question and motivation

- Week 9 asks which part of observed group differences should be interpreted as discrimination.
- This is narrower than Week 8 inequality and harder empirically.
- We need to separate outcome gaps, treatment differences, belief wedges, and equilibrium allocation.

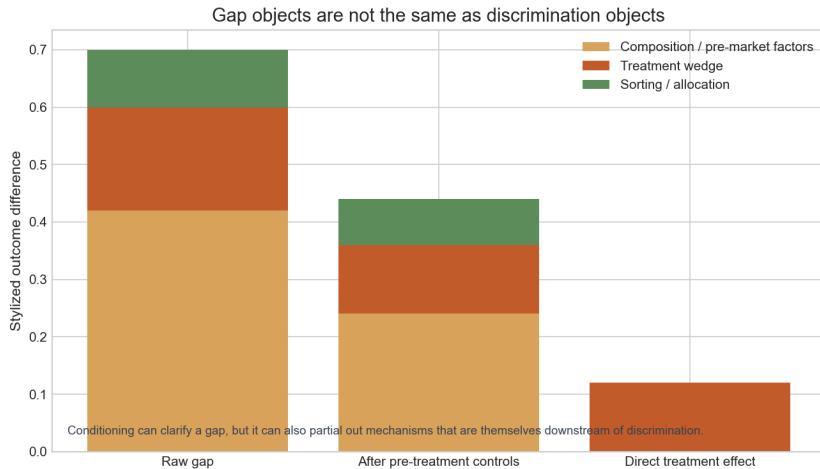
- Week 8 taught that inequality is multi-object: wages, employment, tasks, firms, and amenities can diverge.
- Week 9 asks when group differences inside that architecture reflect discrimination rather than broader dispersion.
- Sorting across firms and segments is part of the discrimination architecture, not just background context.

What is a discrimination object?

$$\Delta = \mathbb{E}[Y \mid G = 1] - \mathbb{E}[Y \mid G = 0], \quad \Delta(x) = \mathbb{E}[Y \mid G = 1, X = x] - \mathbb{E}[Y \mid G = 0, X = x]$$

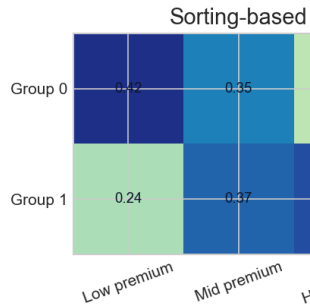
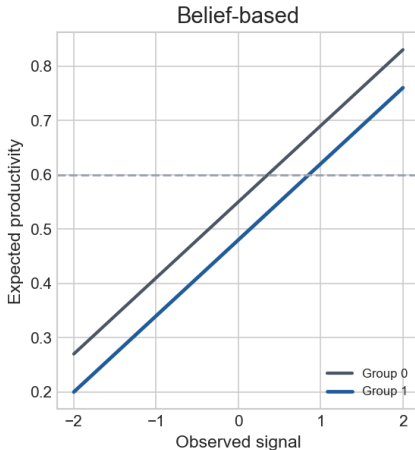
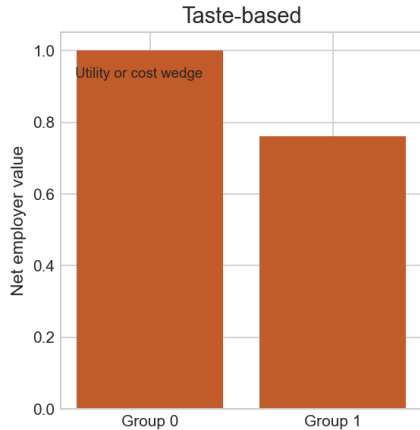
- A raw gap is an outcome difference, not yet a discrimination estimate.
- A conditional gap asks a different question that depends on the control set.
- The key danger is controlling away mechanisms that are themselves downstream of discrimination.

Gap versus discrimination



- Taste-based discrimination: group identity enters utility or effective labor cost.
- Statistical discrimination: noisy signals lead employers to use group-conditioned beliefs.
- Sorting or segmentation: groups are differentially assigned across firms, tasks, occupations, or places.
- Similar reduced-form gaps can emerge from very different mechanisms.

Mechanism comparison



Taste-based discrimination

$$\pi_j = p \cdot q_{ij} - w_i - d_j \mathbf{1}\{G_i = 1\}$$

- A prejudiced employer behaves as if some workers are more costly.
- The wedge can affect hiring, wages, tasks, and retention.
- Competition need not eliminate the wedge when search frictions, customer tastes, or concentration matter.

Hire if $\mathbb{E}[a_i \mid s_i, G_i] \geq c_j$

- Group-conditioned beliefs matter when productivity is imperfectly observed.
- The mechanism operates through beliefs, not necessarily prejudice.
- Anticipated screening can feed back into schooling, training, and task acquisition.

Decomposition logic and bad controls

$$\mathbb{E}[Y \mid G = 1] - \mathbb{E}[Y \mid G = 0] = (\mathbb{E}[X \mid G = 1] - \mathbb{E}[X \mid G = 0])\beta_0 + \mathbb{E}[X \mid G = 1](\beta_1 - \beta_0)$$

- Oaxaca–Blinder style decompositions are descriptive accounting tools.
- The unexplained term is not automatically a causal measure of discrimination.
- Education, occupation, task, or firm assignment may be bad controls if they are downstream of discrimination.

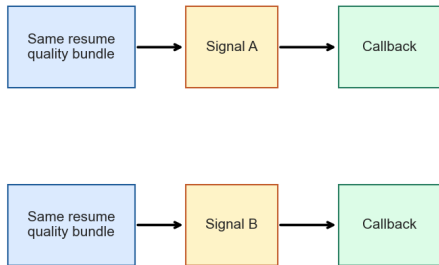
Audit and correspondence identification

$$\tau^{cb} = \mathbb{E}[C_i(1) - C_i(0)]$$

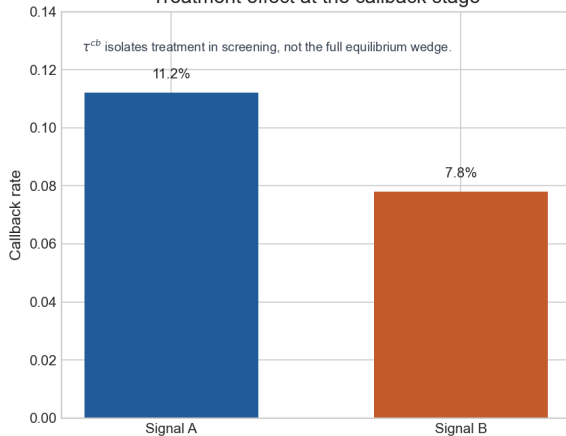
- Correspondence designs hold qualifications fixed while changing signaled identity.
- The estimand is clean at the callback or contact margin.
- It does not by itself recover wage-setting, promotion, or full equilibrium consequences.

Audit-study schematic

Hold qualifications fixed



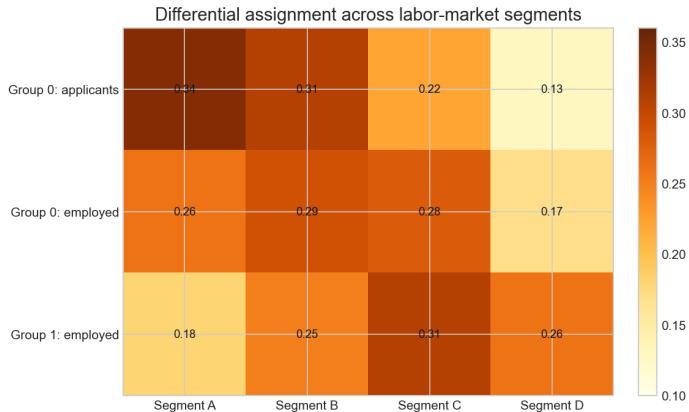
Treatment effect at the callback stage



$$s_{gf} = \Pr(F_i = f \mid G_i = g)$$

- Differential assignment across firms, occupations, or tasks is itself a discrimination object.
- Small within-firm wage gaps can coexist with large between-firm allocation gaps.
- This is the direct bridge from Week 8 firm inequality to Week 10 mobility.

Sorting heatmap



Interpret the cells as assignment shares to firm, occupation, or task segments with different premia or advancement paths.

Firm report cards and frontier measurement

- Modern work asks how discrimination varies across employers, not only on average.
- Report-card designs combine treatment variation with employer-level heterogeneity.
- The frontier challenge is precision: noisy employer rankings can be misleading without shrinkage and uncertainty discipline.

- Week 10 mobility asks how workers move across places, jobs, and generations.
- Week 9 shows that the opportunity set they move through may already be segmented.
- Mobility can therefore be adjustment, escape, or persistence inside a discriminatory allocation system.

- A raw gap is not yet a discrimination estimate.
- Taste-based, statistical, and sorting mechanisms can produce similar reduced-form patterns.
- Identification depends on the margin: decomposition, audit, and equilibrium designs answer different questions.
- Week 9 is the bridge from inequality architecture to segmented mobility and policy design.

Appendix: optional extension block

- Employer report cards, ranking uncertainty, and firm heterogeneity.
- Task-based discrimination and occupational assignment.
- Algorithmic screening, monopsony, and discriminatory market power.